

included in this design to prevent unintentional triggers. However, this physical switch can be replaced with the inbuilt touch sensor in the code.

The GSM and GPS modules can connect to any IndusBoard pin. All pins are flexible and can be configured in the code.

Software

The library for the sensor used here is compatible with the IndusBoard, which has an inbuilt LSM303 sensor for accelerometer, magnetometer, and temperature data. To utilise this, install the STM LSM303 library. The Adafruit GPS library is used for GPS functionality.

After installing these libraries, download the code from electronicsforu.com and upload it to the device.

The code defines the pins and variables for the GPS, GSM, buzzer, and other components first. The acceleration, temperature, and GPS data are checked and updated within the loop.

If the acceleration threshold is exceeded, the device sends an SOS message containing the location and activates the buzzer for alerts. The system also checks for a push button press, sending an alert if triggered.

The phone number can be customised by editing the code. Fig. 6 shows the snippet of source code.

Testing

Once the code is uploaded and the device is assembled, it can be powered using the 3.3V or 3.7V battery. Pressing the SOS button sends a message to the phone number containing the location and activates the buzzer sound for an alert. The same response occurs when the acceleration threshold detects a signal, indicating a sudden fall or accident. SOS messages with location data are displayed on the right-hand side of the prototype (see Fig. 2).

Functions of the device

Emergency activation. The device activates emergency protocols when the user presses the push button. The buzzer sounds to alert those nearby to an emergency situation. The GPS module retrieves the current location coordinates (latitude and longitude). The GSM module sends an SMS containing the GPS coordinates and any relevant sensor data (temperature readings) to pre-stored emergency contacts.

Automatic alerts. The device automatically sends alerts without user intervention if the MEMS sensor detects a fall or the temperature exceeds predefined thresholds. It sends multiple SMS notifications to ensure that contacts receive the alert. It activates the buzzer for audible notifications.

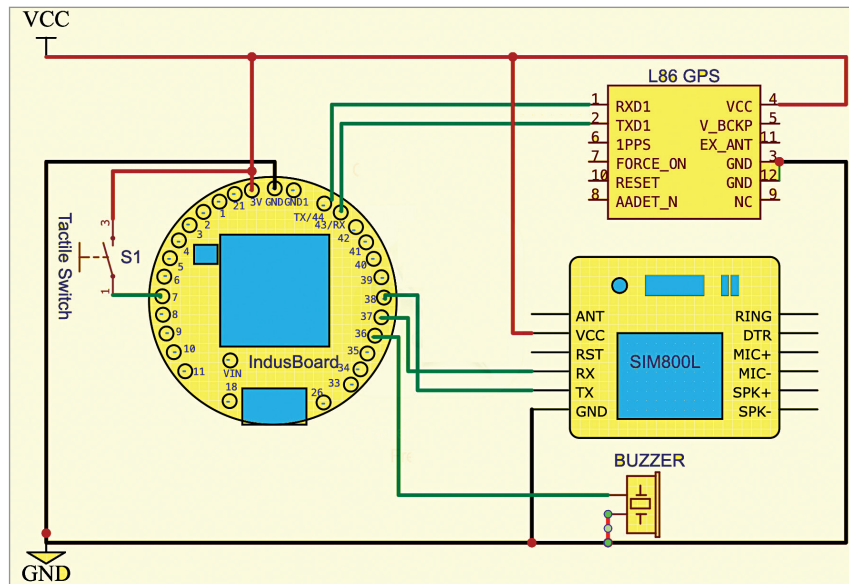


Fig. 3: Circuit diagram

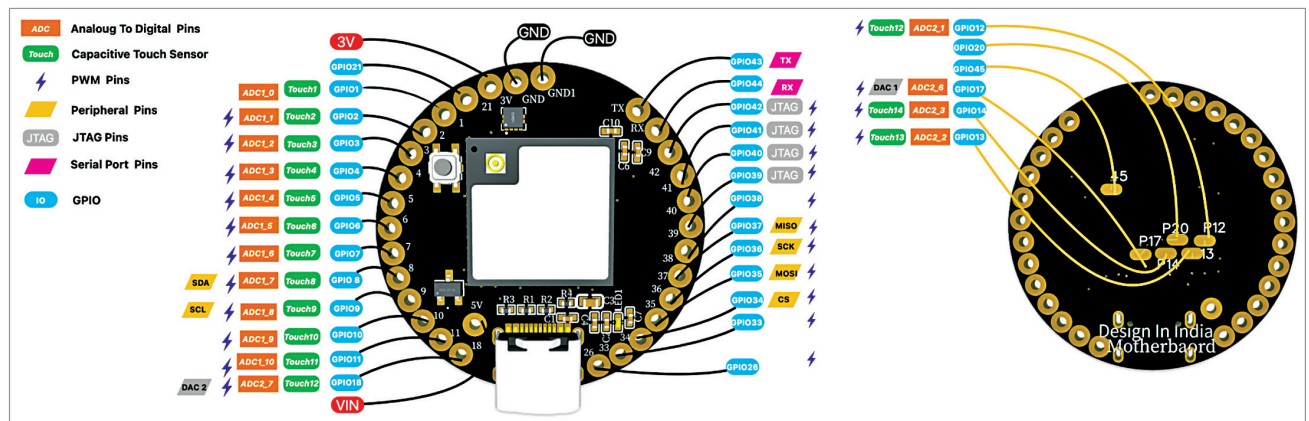


Fig. 4: IndusBoard pinouts